

This by no means implies or excludes similar questions or marks distribution on the exam.

1. (6 marks) Design a Finite State Machine that recognizes all binary strings that **end with 1010** and have **an odd number of 0's**.

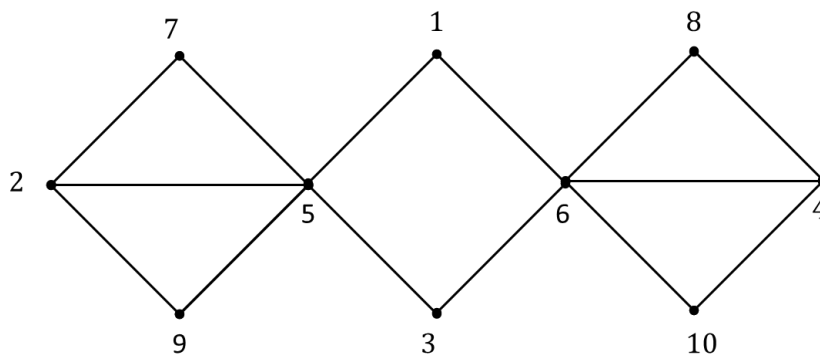
Marks will be taken away for missing details; clearly label all states. For the full marks, your FSM must have the minimum number of required states.

2. (5 marks) A *modified* depth first search (dfs) is shown below. Suppose **dfs(1)** is called using the graph shown. You must:

- Show the contents of the return stack
- Show the resulting screen output

```
dfs (vertex v):
    for each neighbor w of v:      //in increasing order
        if w is unvisited:
            visit w
            print w
            dfs (w)
        else:
            print v
```

RETURN STACK:

[illegible]

PRINTED ON THE SCREEN:

[illegible]

3. (3 marks) Given

$$\sum_{k=1}^{100} (2a_{k+1} + 3k) = 19250 \quad \text{and} \quad \sum_{j=1}^{101} (a_j - 4) = 1745$$

Find a_1

4. (6 marks total) Consider the text message below.

I WILL LACK CHILLIWACK CHILL

- a. (5 marks) Determine the character frequencies (i.e., counts) for this text. Then use the Huffman algorithm to produce an optimal tree for encoding this text and write the codewords in the table. Show all steps.

Character	Frequency	Codeword
<space>		
A		
C		
H		
I		
K		
L		
W		

*** There is lots of space on the next page if needed. Part b) is also on the next page ***

- b. (1 mark) How many bits are needed in total to encode the text message using the Huffman codewords you obtained?

5. (6 marks total)

- (a) (3 marks) We define “LLVR_Order” tree Traversal Algorithm as follows:

LLVR_Order Traversal Algorithm:

- Step 1: (start)
Go to the root.
- Step 2: (go **left**)
Go to the left subtree, if one exists, and do a LLVR_Order traversal.
- Step 3: (go **left**)
Go to the left subtree, if one exists, and do a LLVR_Order traversal.
- Step 4: (**visit**)
Visit the root.
- Step 5: (go **right**)
Go to the right subtree, if one exists, and do a LLVR_Order traversal.

The vertices of a binary tree T are visited in the following order:

Y G Y G U Y G Y G U F S W W Q P

Recover (draw) the tree T.

- (b) (3 marks) A tree with 80 vertices has:
- 6 vertices with degree 5,
 - x vertices with degree 4,
 - 9 vertices with degree 3,
 - y vertices with degree 1,

Determine the values of x and y.

6. (6 marks) Find the value of the variable `time` (in terms of `n`) after the following programming segment is executed. Your formula must be in the simplest form, fully expanded and terms combined.
(Eg. $time = 7n^2 + 4n + 9$)

****Note that the two *for loops* are *nested loops***

```
time = 350;
for (i = 5; i <= n; i++){
    time = time + 10i;
    for (j = 1; j <= i - 1; j++){
        time = time + 24j + 6i;
    }
}
// assume n>6
```

*****In order to receive marks your approach must rely on sigma notation, summation properties and usage of given formula(s). Answers based on combinatorics will not be accepted.**

Use the following formulas: $\sum_{i=1}^n i = \frac{n \cdot (n+1)}{2}$ $\sum_{i=1}^n i^2 = \frac{n \cdot (n+1) \cdot (2n+1)}{6}$

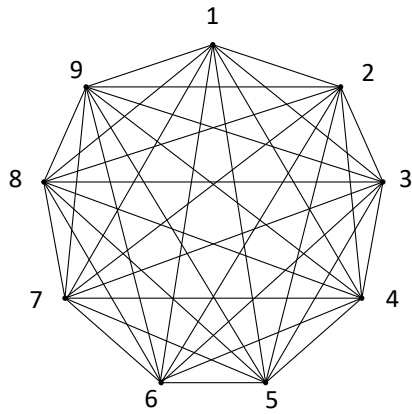
7. (4 marks) For input n , the following function returns $-2n + 17 \cdot 3^{2n+1} - 9$. Prove the correctness of this formula using mathematical induction.

MysteryFunction (positive integer n)

```
i = 1
j = 42
while i <= n do
    i = i + 1
    j = 9j + 16i + 38
end while

return j
```


8. (4 marks) Consider the graph below that has an edge between any two vertices.



Find the number of distinct paths between vertices 8 and 3.

Recall that no vertex can repeat in a path.